

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250276568

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Balding; Andrew Geoffrey et al.

BATTERY SYSTEM MOUNTING APPARATUS FOR A VEHICLE

Abstract

An apparatus for affixing a battery system component to a vehicle may include a mounting plate configured to be removably affixed to the vehicle. The battery system component is affixed to the mounting plate. The apparatus further may include a mounting plate hanger affixed to the mounting plate. The mounting plate hanger is configured to be removably affixed to the vehicle.

Inventors: Balding; Andrew Geoffrey (Wixom, MI), Creffield; Damion (Manchester, MI)

Applicant: GM Global Technology Operations LLC (Detroit, MI)

Family ID: 96738074

Appl. No.: 18/592770

Filed: March 01, 2024

Publication Classification

Int. Cl.: B60K1/04 (20190101); B60R16/03 (20060101); B60R16/08 (20060101); B62D25/20 (20060101); B62D25/24 (20060101); H01M50/209 (20210101); H01M50/24 (20210101); H01M50/249 (20210101); H01M50/262 (20210101)

U.S. Cl.:

CPC B60K1/04 (20130101); B60R16/03 (20130101); B60R16/08 (20130101); B62D25/20 (20130101); B62D25/24 (20130101); H01M50/209 (20210101); H01M50/24 (20210101); H01M50/249 (20210101); H01M50/262 (20210101); B60K2001/0438 (20130101); H01M2220/20 (20130101)

Background/Summary

INTRODUCTION

[0001] The present disclosure relates to an apparatus and system for mounting battery system components to a vehicle.

[0002] To increase performance and ease of manufacturing, battery powered electric vehicles and hybrid electric vehicles may be equipped with battery storage systems which are configured to provide electrical energy to power electromechanical components and systems of the vehicle. Battery storage systems may be packaged in a sealed container to provide structure and environmental/electrical isolation for the battery system. However, current battery storage systems may be large and heavy, requiring specialized equipment for installation to the vehicle during vehicle manufacturing. Additionally, current battery storage systems may require removal of the entire storage system from the vehicle in order to perform service and/or maintenance of the battery system. Furthermore, current battery system packaging apparatuses may require significant modification to vehicle structure in order to accommodate the large and/or heavy container.

[0003] Thus, while current battery packaging systems and apparatuses achieve their intended purpose, there is a need for a new and improved system and apparatus for affixing a battery system component to a vehicle.

SUMMARY

[0004] According to several aspects, an apparatus for affixing a battery system component to a vehicle is provided. The apparatus may include a mounting plate configured to be removably affixed to the vehicle. The battery system component is affixed to the mounting plate. The apparatus further may include a mounting plate hanger affixed to the mounting plate. The mounting plate hanger is configured to be removably affixed to the vehicle.

[0005] In another aspect of the present disclosure, the mounting plate further may include one or more attachment features configured to be affixed to the vehicle using a removable fastener.

[0006] In another aspect of the present disclosure, the one or more attachment features are affixed to a vehicle body cross member of the vehicle using the removable fastener.

[0007] In another aspect of the present disclosure, the battery system component further may include a battery module including one or more battery cells.

[0008] In another aspect of the present disclosure, the battery system component further may include a battery disconnect unit (BDU).

[0009] In another aspect of the present disclosure, the mounting plate hanger further may include a rigid support member affixed to the mounting plate and removably affixed to the vehicle.

[0010] In another aspect of the present disclosure, the rigid support member further may include a first end affixed to the mounting plate. The rigid support member further may include a second end affixed to the vehicle using a removable fastener. The second end is opposite the first end.

[0011] In another aspect of the present disclosure, the rigid support member further may include a trapezoidal rigid support member. The first end of the trapezoidal rigid support member is affixed to the mounting plate and the second end of the trapezoidal rigid support member is affixed to a vehicle body floor pan of the vehicle using a removable fastener.

[0012] In another aspect of the present disclosure, the vehicle body floor pan further may include an access panel providing access to the battery system component.

[0013] In another aspect of the present disclosure, the apparatus further may include a liner disposed between the battery system component and the vehicle. The liner is configured to isolate the battery system component from the vehicle.

[0014] According to several aspects, a system for affixing a battery system component to a vehicle is provided. The system may include a vehicle body including a vehicle body floor pan having a first surface and a second surface. The first surface is parallel to the second surface. The vehicle body further may include a plurality of vehicle body cross members, each of the plurality of vehicle body cross members having a first end, a second end, and a first surface. The first end of each of

the plurality of the vehicle body cross members is affixed to the second surface of the vehicle body floor pan such that the first surface of each of the plurality of vehicle body cross members is substantially perpendicular to the second surface of the vehicle body floor pan. The system further may include a mounting module removably affixed between two of the plurality of vehicle body cross members, the mounting module including a mounting plate having a first surface. The mounting module further may include a mounting plate hanger having a first end and a second end. The first end is affixed to the first surface of the mounting plate. The second end is configured to be removably affixed to the second surface of the vehicle body floor pan. The system further may include the battery system component affixed to the first surface of the mounting plate.

[0015] In another aspect of the present disclosure, each of the plurality of vehicle body cross members further may include one or more penetrations in the first surface of each of the plurality of vehicle body cross members. Each of the one or more penetrations allows for passage of one or more electrical or fluid conductors for electrical or fluid connection to the battery system component.

[0016] In another aspect of the present disclosure, the vehicle body floor pan further may include one or more access panels disposed on the first surface of the vehicle body floor pan. The one or more access panels provide access to the one or more electrical or fluid conductors.

[0017] In another aspect of the present disclosure, the mounting plate further may include one or more attachment features configured to be affixed to the second end of one or more of the plurality of vehicle body cross members using removable fasteners.

[0018] In another aspect of the present disclosure, the mounting plate hanger further may include a trapezoidal rigid support member. A first end of the trapezoidal rigid support member is affixed to the first surface of the mounting plate and a second end of the trapezoidal rigid support member is affixed to the second surface of the vehicle body floor pan using a removable fastener.

[0019] In another aspect of the present disclosure, the battery system component further may include at least one of: (i) a battery module including one or more battery cells, and (ii) a battery disconnect unit (BDU).

[0020] In another aspect of the present disclosure, the system further may include a liner disposed between the battery system component and the vehicle body. The liner is configured to isolate the battery system component from the vehicle body.

[0021] According to several aspects, an apparatus for affixing a battery system component to a vehicle is provided. The apparatus may include a mounting module including a mounting plate having a first surface. The battery system component is affixed to the first surface. The apparatus further may include a mounting plate hanger having a first end and a second end. The first end is affixed to the first surface of the mounting plate. The second end is configured to be removably affixed to a vehicle body floor pan.

[0022] In another aspect of the present disclosure, the mounting plate hanger further may include a trapezoidal rigid support member. A first end of the trapezoidal rigid support member is affixed to the first surface of the mounting plate and a second end of the trapezoidal rigid support member is affixed to the vehicle body floor pan using a removable fastener.

[0023] In another aspect of the present disclosure, the mounting module further may include at least two mounting plate hangers affixed to the first surface of the mounting plate. The battery system component is disposed between the at least two mounting plate hangers on the first surface of the mounting plate.

[0024] Further areas of applicability will become apparent from the description provided herein. It should be understood that the description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] The drawings described herein are for illustration purposes only and are not intended to limit the scope of the present disclosure in any way.

[0026] FIG. **1** is a schematic diagram of a system for affixing a battery system component to a vehicle, according to an exemplary embodiment;

[0027] FIG. **2** is an exploded perspective view of the system for affixing a battery system component to a vehicle, according to an exemplary embodiment;

[0028] FIG. **3** is an assembled perspective view of the system for affixing a battery system component to a vehicle, according to an exemplary embodiment;

[0029] FIG. **4** is an assembled bottom perspective view of the system for affixing a battery system component to a vehicle, according to an exemplary embodiment;

[0030] FIG. **5** is a perspective view of a mounting module with a first exemplary battery system component, according to an exemplary embodiment;

[0031] FIG. **6** is a perspective view of a mounting module with a second exemplary battery system component, according to an exemplary embodiment;

[0032] FIG. **7** is a top-down perspective view of a portion of the system for affixing a battery system component to a vehicle, according to an exemplary embodiment; and

[0033] FIG. **8** is a magnified view of the assembled perspective view of FIG. **3**, according to an exemplary embodiment.

DETAILED DESCRIPTION

[0034] The following description is merely exemplary in nature and is not intended to limit the present disclosure, application, or uses.

[0035] In aspects of the present disclosure, a new and improved battery storage system for battery electric vehicles (BEVs), plug-in hybrid electric vehicles (PHEVs), and/or hybrid electric vehicles (HEVs) is provided. The battery storage system is packaged to provide structure and environmental/electrical isolation. Advantageously, the new and improved battery storage system of the present disclosure adds less weight to the vehicle. This is because the new and improved battery storage system of the present disclosure utilizes existing vehicle structure to provide structure to the battery storage system. This is in contrast to conventional sealed battery containers that are larger and heavier, requiring specialized equipment and skill for installation during vehicle assembly and removal/re-installation during maintenance. Furthermore, use of sealed battery containers may require re-design of vehicle body structure in order to accommodate the size, shape, and weight of such sealed battery containers. Accordingly, the present disclosure provides a new and improved battery storage system having a modular design for affixing batteries and battery system components to a vehicle.

[0036] Referring to FIG. **1**, a schematic diagram of a system for affixing a battery system component to a vehicle is illustrated and generally indicated by reference number **10**. The system **10** is shown with a cutaway view of an exemplary vehicle **12**. While a passenger vehicle is illustrated, it should be appreciated that the vehicle **12** may be any type of vehicle without departing from the scope of the present disclosure. In an exemplary embodiment, the system **10** is disposed on an underside of the vehicle **12**. More particularly, the system **10** is integrated with a body of the vehicle **12**, as will be discussed in greater detail below.

[0037] Referring now to FIGS. **2-4**, an exploded perspective view of the system **10** is shown in FIG. **2**, an assembled perspective view of the system **10** is shown in FIG. **3**, and an assembled bottom perspective view of the system **10** is shown in FIG. **4**. The system **10** generally includes a vehicle body **14**, a mounting module **16**, a battery system component **18**, and a liner **20**. While FIGS. **2-4** show an embodiment including three mounting modules **16**, three battery system components **18**, and three liners **20**, it should be understood that embodiments including any number of mounting modules **16**, battery system components **18**, and liners **20** are within the scope

of the present disclosure. Furthermore, the system **10** is shown with a plurality of fasteners **22**. In the scope of the present disclosure, the plurality of fasteners **22** includes removable fasteners, such as, for example, screws, bolts, and/or the like. The plurality of fasteners **22** are configured to engage with nuts, threads, and/or the like to form a removable affixture between two objects. It should be understood that the location, type, and size of the plurality of fasteners **22** shown in FIGS. 2-4 are merely exemplary in nature, and that FIGS. 2-4 are not necessarily to scale.

[0038] The vehicle body **14** includes a vehicle body floor pan **24** and a plurality of vehicle body cross members **26**. The vehicle body floor pan **24** provides structure to the vehicle body **14** and provides a floor surface for an interior of the vehicle **12**. Interior components of the vehicle **12** (e.g., seats of the vehicle **12**) may be mounted to the vehicle body floor pan **24**. In an exemplary embodiment, the vehicle floor pan **24** has a first surface **24a** and a second surface **24b**. The first surface **24a** of the vehicle body floor pan **24** faces the interior of the vehicle **12** and the second surface **24b** faces an exterior of the vehicle **12**. In an exemplary embodiment, the vehicle body floor pan **24** is penetrated by a plurality of access ports **28a** allowing access from the first surface **24a** (i.e., the interior of the vehicle **12**) to the battery system component **18**, as will be discussed in greater detail below. The vehicle body floor pan **24** further may include a plurality of access panels **28b** disposed on the first surface **24a** to cover the plurality of access ports **28a**. In a non-limiting example, the plurality of access panels **28b** are removably affixed with the fasteners **22**. In an exemplary embodiment, the fasteners **22** holding the plurality of access panels **28b** may be removed from the first surface **24a** (i.e., the interior of the vehicle **12**). In another exemplary embodiment, the fasteners **22** holding the plurality of access panels **28b** may be removed from the second surface **24b** (i.e., the exterior of the vehicle **12**). It should be understood that the quantity, size, shape, and location of the plurality of the access ports **28a** and the access panels **28b** shown in FIGS. 2-4 are merely exemplary in nature, and that FIGS. 2-4 are not necessarily to scale. In a non-limiting example, the vehicle body floor pan **24** is made of a sheet metal and/or composite material (e.g., steel, aluminum, carbon fiber, and/or the like).

[0039] The plurality of vehicle body cross members **26** provide additional structural support to the vehicle body floor pan **24**. In an exemplary embodiment, each vehicle body cross member **26** has a first end **30a**, a second end **30b**, and a first surface **32**. In a non-limiting example, the first end **30a** of each vehicle body cross member **26** is affixed to the second surface **24b** of the vehicle body floor pan **24** such that the first surface **32** of each vehicle body cross member **26** is substantially perpendicular to the second surface **24b** of the vehicle body floor pan **24**. In the scope of the present disclosure, substantially perpendicular means that an angle measured between the first surface **32** of each vehicle body cross member **26** and the second surface **24b** of the vehicle body floor pan **24** is approximately ninety degrees (90°) within a predetermined tolerance (e.g., $\pm 10\%$).

[0040] In a non-limiting example, the first end **30a** is affixed to the second surface **24b** of the vehicle body floor pan **24** using a weld, an angled bracket and fasteners, and/or the like. In an exemplary embodiment, each vehicle body cross member **26** has one or more penetrations **34** in at least the first surface **32** of each vehicle body cross member **26** allowing for connection to the battery system component **18**, as will be discussed in greater detail below. It should be understood that the quantity, size, shape, and location of the one or more penetrations **34** shown in FIGS. 2-4 are merely exemplary in nature, and that FIGS. 2-4 are not necessarily to scale. It should be understood that the quantity, size, shape, and location of the plurality of vehicle body cross members **26** shown in FIGS. 2-4 are merely exemplary in nature. In a non-limiting example, the plurality of vehicle body cross members **26** are made of a sheet metal and/or composite material (e.g., steel, aluminum, carbon fiber, and/or the like).

[0041] Referring to FIG. 5, a perspective view of the mounting module **16** with a first exemplary battery system component **18a** is shown. The mounting module **16** is used to contain the battery system component **18** (e.g., the first exemplary battery system component **18a**) and affix the battery system component **18** to the vehicle body **14**. In an exemplary embodiment, the mounting

module **16** includes a mounting plate **40** and a mounting plate hanger **42**. In an exemplary embodiment, the mounting plate **40** has at least a first surface **40a** and one or more attachment features **44**. The one or more attachment features **44** are configured to be attached to the second end **30b** of the vehicle body cross members **26** using the fasteners **22**. In an exemplary embodiment, the one or more attachment features **44** are through-holes or threaded holes in at least the first surface **40a** of the mounting plate **40** disposed proximally to one or more edges or corners of the mounting plate **40**. It should be understood that the quantity, size, shape, and location of the one or more attachment features **44** shown in FIG. 5 are merely exemplary in nature, and that FIG. 5 is not necessarily to scale. In a non-limiting example, the mounting plate **40** is made of a sheet metal and/or composite material (e.g., steel, aluminum, carbon fiber, and/or the like).

[0042] The mounting plate hanger **42** is used to provide additional structural support and affixture of the mounting plate **40** to the vehicle body **14**. In an exemplary embodiment, the mounting plate hanger **42** has at least a first end **46a** and a second end **46b**. The first end **46a** of the mounting plate hanger **42** is affixed to the first surface **40a** of the mounting plate **40**. In a non-limiting example, the first end **46a** is welded to the first surface **40a** of the mounting plate **40**. In another non-limiting example, the first end **46a** is affixed to the first surface **40a** of the mounting plate **40** with fasteners. The second end **46b** of the mounting plate hanger **42** includes a through-hole, a threaded hole, a captive nut, and/or the like for removable affixture of the second end **46b** to the second surface **24b** of the vehicle body floor pan **24** using fasteners **22**. In an exemplary embodiment, the mounting plate hanger **42** is a rigid support member made of a metal and/or composite material (e.g., steel, aluminum, carbon fiber, and/or the like) sheet, extrusion, tube, box, and/or the like. In a non-limiting example, the mounting plate hanger **42** is a trapezoidal shaped rigid support member. In an exemplary embodiment, one or more mounting plate hangers **42** are affixed proximally to one or more edges of the mounting plate **40**. It should be understood that the quantity, size, shape, and location of the mounting plate hangers **42** shown in FIG. 5 are merely exemplary in nature, and that FIG. 5 is not necessarily to scale. In some embodiments, additional mounting plate hangers **42** are included depending on the needs of a particular application, battery system component **18**, and/or the like.

[0043] With continued reference to FIG. 5, the first exemplary battery system component **18a** is shown with the mounting module **16**. In an exemplary embodiment, the first exemplary battery system component **18a** includes a battery module **48** including one or more battery cells **50**. In a non-limiting example, the battery module **48** stores and provides electrical energy in the form of direct current (DC). In an exemplary embodiment, the one or more battery cells **50** (e.g., lithium-ion battery cells) are electrically connected in series and/or parallel to provide an increased voltage and/or current-carrying capacity. It should be understood that the one or more battery cells **50** may include any chemistry and form-factor of battery cells, including, for example, cylindrical battery cells.

[0044] The first exemplary battery system component **18a** further includes one or more electrical or fluid conductors **52** for interconnection (i.e., electrical and/or fluid connection) between adjacent battery system components **18** (as shown in FIG. 3). In a non-limiting example, the one or more electrical or fluid conductors **52** include, for example, electrical bus bars, electrical cables, coolant lines, air ducts, gas venting ducts, and/or the like. The one or more electrical or fluid conductors **52** pass through the one or more penetrations **34** to provide an electrical and/or fluid connection between adjacent battery system components **18**. In an exemplary embodiment, the one or more electrical or fluid conductors **52** may be assembled from one or more of the plurality of access ports **28a**. In another exemplary embodiment, the one or more electrical or fluid conductors **52** may be assembled from the exterior (i.e., an underside) of the vehicle **12**. It should be understood that the type, quantity, size, shape, routing, and location of the one or more electrical or fluid conductors **52** are merely exemplary in nature, and that FIG. 5 is not necessarily to scale.

[0045] Referring to FIG. 6, a perspective view of the mounting module **16** as described above in

reference to FIG. 5 is shown with a second exemplary battery system component **18b**. In an exemplary embodiment, the second exemplary battery system component **18b** is a battery disconnect unit (BDU) **54**.

[0046] The BDU **54** is used to control the connection and disconnection of one or more battery modules **48** to other systems of the vehicle **12**. In an exemplary embodiment, the BDU **54** includes a BDU controller, a BDU sensor, and a BDU switch. In a non-limiting example, the BDU controller uses the BDU sensor to monitor battery system metrics (e.g., voltage, current, temperature, and/or the like) and detect adverse conditions (e.g., overvoltage or overcurrent conditions, thermal runaway, and/or the like). In response to detecting adverse conditions, the BDU controller actuates the BDU switch to isolate the battery system from other systems of the vehicle **12**. In another non-limiting example, the BDU controller receives input from an external device (e.g., an emergency shutoff switch, a wired/wireless interface, and/or the like) and actuates the BDU switch to isolate the battery system based at least in part on the input from the external device.

[0047] The second exemplary battery system component **18b** further includes the one or more electrical or fluid conductors **52** for interconnection (i.e., electrical and/or fluid connection) between adjacent battery system components **18**. In a non-limiting example, the one or more electrical or fluid conductors **52** include, for example, electrical bus bars, electrical cables, coolant lines, air ducts, gas venting ducts, and/or the like. The one or more electrical or fluid conductors **52** pass through the one or more penetrations **34** to provide an electrical and/or fluid connection between adjacent battery system components **18**. It should be understood that the type, quantity, size, shape, routing, and location of the one or more electrical or fluid conductors **52** are merely exemplary in nature, and that FIG. 6 is not necessarily to scale.

[0048] Referring to FIG. 7, a top perspective view of a portion of the floor pan **24** is shown. The top perspective view is shown with the access panel **28b** removed, illustrating access to the one or more electrical or fluid conductors **52** through the access port **28a**. The one or more electrical or fluid conductors **52** may be accessed using the access port **28a** for service, maintenance, testing, assembly/disassembly, and/or the like.

[0049] Referring again to FIGS. 2-6, in the scope of the present disclosure, the battery system component **18** is any component of an electrical system of the vehicle **12**, and more particularly, any component of an energy storage system of the vehicle **12**. Two examples of battery system components **18** are discussed above in reference to FIGS. 5 and 6 (i.e., the first exemplary battery system component **18a** and the second exemplary battery system component **18b**). In general, the battery system component **18** may include any component of the electrical system of the vehicle **12**, including but not limited to: battery modules, battery cells, BDUs, power electronics (e.g., fuses, contactors, converters, inverters, etc.), battery management systems (BMS), battery cooling system components (e.g., pumps, hoses, radiators, coolant reservoirs, etc.), battery heating system components, battery safety system components (e.g., fire suppression equipment, thermal runaway prevention/mitigation equipment, etc.), and/or the like. The battery system component **18** is affixed to the first surface **40a** of the mounting plate **40**. In a non-limiting example, the battery system component **18** is affixed to the mounting plate **40** using one or more brackets, fasteners, and/or the like.

[0050] Referring to FIG. 8, a magnified view **56** of the assembled perspective view of FIG. 3 is shown. Referring again to FIGS. 2-3 and 8, the liner **20** is used to isolate the battery system component **18** and/or the mounting module **16** from the vehicle body **14** and contain and/or direct gases vented from the battery system component **18**. The liner **20** is configured to provide one or more of: electrical, thermal, chemical, environmental, and physical isolation and/or protection of the battery system component **18**. In an exemplary embodiment, the liner **20** is disposed over the battery system component **18**, providing a barrier between the battery system component **18** and the vehicle body **14** when the system **10** is assembled. In another exemplary embodiment, the liner

20 is a coating which is disposed on the second surface **24b** of the vehicle body floor pan **24** and one or more surfaces of the plurality of vehicle body cross members **26**. In some examples, the liner **20** includes penetrations for the one or more electrical or fluid conductors **52** and/or the plurality of access ports **28a**. In a non-limiting example, the liner **20** is a blow-molded polymer structure. In another non-limiting example, the liner **20** is a curable coating which adheres to the vehicle body **14** and cures after application to the vehicle body **14**. It should be understood that the size, shape, and location of the liner **20** are merely exemplary in nature.

[0051] The system **10** and mounting module **16** of the present disclosure offer several advantages. By utilizing existing vehicle structure such as the vehicle body **14**, to provide structure for the battery system components **18**, the need for additional material and thus added weight due to structural components is reduced. Furthermore, by providing the mounting module **16** as a modular apparatus, installation and maintenance may be performed on individual mounting modules **16**, reducing the need for handling of large and/or heavy components. The access ports **28a** and access panels **28b** allow for servicing of electrical or fluid conductors **52** without removal of mounting modules **16**. The liner **20** isolates and protects the battery system components **18**. Due to the modular design of the system **10**, the present disclosure is adaptable, scalable, and applicable for use with many types of battery system components and many vehicle body designs. For example, the system **10** may be scaled to provide a specific amount of battery system energy storage based on vehicle size, customer needs, and/or the like. Furthermore, the system **10** is adaptable for use with various current and future battery chemistries and power system technologies and topologies.

[0052] The description of the present disclosure is merely exemplary in nature and variations that do not depart from the gist of the present disclosure are intended to be within the scope of the present disclosure. Such variations are not to be regarded as a departure from the spirit and scope of the present disclosure.

Claims

1. An apparatus for affixing a battery system component to a vehicle, the apparatus comprising: a mounting plate configured to be removably affixed to the vehicle, wherein the battery system component is affixed to the mounting plate; and a mounting plate hanger affixed to the mounting plate, wherein the mounting plate hanger is configured to be removably affixed to the vehicle.
2. The apparatus of claim 1, wherein the mounting plate further comprises: one or more attachment features configured to be affixed to the vehicle using a removable fastener.
3. The apparatus of claim 2, wherein the one or more attachment features are affixed to a vehicle body cross member of the vehicle using the removable fastener.
4. The apparatus of claim 1, wherein the battery system component further comprises: a battery module including one or more battery cells.
5. The apparatus of claim 1, wherein the battery system component further comprises: a battery disconnect unit (BDU).
6. The apparatus of claim 1, wherein the mounting plate hanger further comprises: a rigid support member affixed to the mounting plate and removably affixed to the vehicle.
7. The apparatus of claim 6, wherein the rigid support member further comprises: a first end affixed to the mounting plate; and a second end affixed to the vehicle using a removable fastener, wherein the second end is opposite the first end.
8. The apparatus of claim 7, wherein the rigid support member further comprises: a trapezoidal rigid support member, wherein the first end of the trapezoidal rigid support member is affixed to the mounting plate and the second end of the trapezoidal rigid support member is affixed to a vehicle body floor pan of the vehicle using a removable fastener.
9. The apparatus of claim 8, wherein the vehicle body floor pan further comprises: an access panel providing access to the battery system component.

- 10.** The apparatus of claim 1, further comprising: a liner disposed between the battery system component and the vehicle, wherein the liner is configured to isolate the battery system component from the vehicle.
- 11.** A system for affixing a battery system component to a vehicle, the system comprising: a vehicle body including: a vehicle body floor pan having a first surface and a second surface, wherein the first surface is parallel to the second surface; and a plurality of vehicle body cross members, each of the plurality of vehicle body cross members having a first end, a second end, and a first surface, and wherein the first end of each of the plurality of the vehicle body cross members is affixed to the second surface of the vehicle body floor pan such that the first surface of each of the plurality of vehicle body cross members is substantially perpendicular to the second surface of the vehicle body floor pan; a mounting module removably affixed between two of the plurality of vehicle body cross members, the mounting module including: a mounting plate having a first surface; and a mounting plate hanger having a first end and a second end, wherein the first end is affixed to the first surface of the mounting plate, and wherein the second end is configured to be removably affixed to the second surface of the vehicle body floor pan; and the battery system component affixed to the first surface of the mounting plate.
- 12.** The system of claim 11, wherein each of the plurality of vehicle body cross members further comprises: one or more penetrations in the first surface of each of the plurality of vehicle body cross members, wherein each of the one or more penetrations allows for passage of one or more electrical or fluid conductors for electrical or fluid connection to the battery system component.
- 13.** The system of claim 12, wherein the vehicle body floor pan further comprises: one or more access panels disposed on the first surface of the vehicle body floor pan, wherein the one or more access panels provide access to the one or more electrical or fluid conductors.
- 14.** The system of claim 11, wherein the mounting plate further comprises: one or more attachment features configured to be affixed to the second end of one or more of the plurality of vehicle body cross members using removable fasteners.
- 15.** The system of claim 11, wherein the mounting plate hanger further comprises: a trapezoidal rigid support member, wherein a first end of the trapezoidal rigid support member is affixed to the first surface of the mounting plate and a second end of the trapezoidal rigid support member is affixed to the second surface of the vehicle body floor pan using a removable fastener.
- 16.** The system of claim 11, wherein the battery system component further comprises: at least one of: (i) a battery module, wherein the battery module includes one or more battery cells, and (ii) a battery disconnect unit (BDU).
- 17.** The system of claim 11, further comprising: a liner disposed between the battery system component and the vehicle body, wherein the liner is configured to isolate the battery system component from the vehicle body.
- 18.** An apparatus for affixing a battery system component to a vehicle, the apparatus comprising: a mounting module including: a mounting plate having a first surface, wherein the battery system component is affixed to the first surface; and a mounting plate hanger having a first end and a second end, wherein the first end is affixed to the first surface of the mounting plate, and wherein the second end is configured to be removably affixed to a vehicle body floor pan.
- 19.** The apparatus of claim 18, wherein the mounting plate hanger further comprises: a trapezoidal rigid support member, wherein a first end of the trapezoidal rigid support member is affixed to the first surface of the mounting plate and a second end of the trapezoidal rigid support member is affixed to the vehicle body floor pan using a removable fastener.
- 20.** The apparatus of claim 19, wherein the mounting module further comprises: at least two mounting plate hangers affixed to the first surface of the mounting plate, wherein the battery system component is disposed between the at least two mounting plate hangers on the first surface of the mounting plate.
-

